TVS MOBILITY

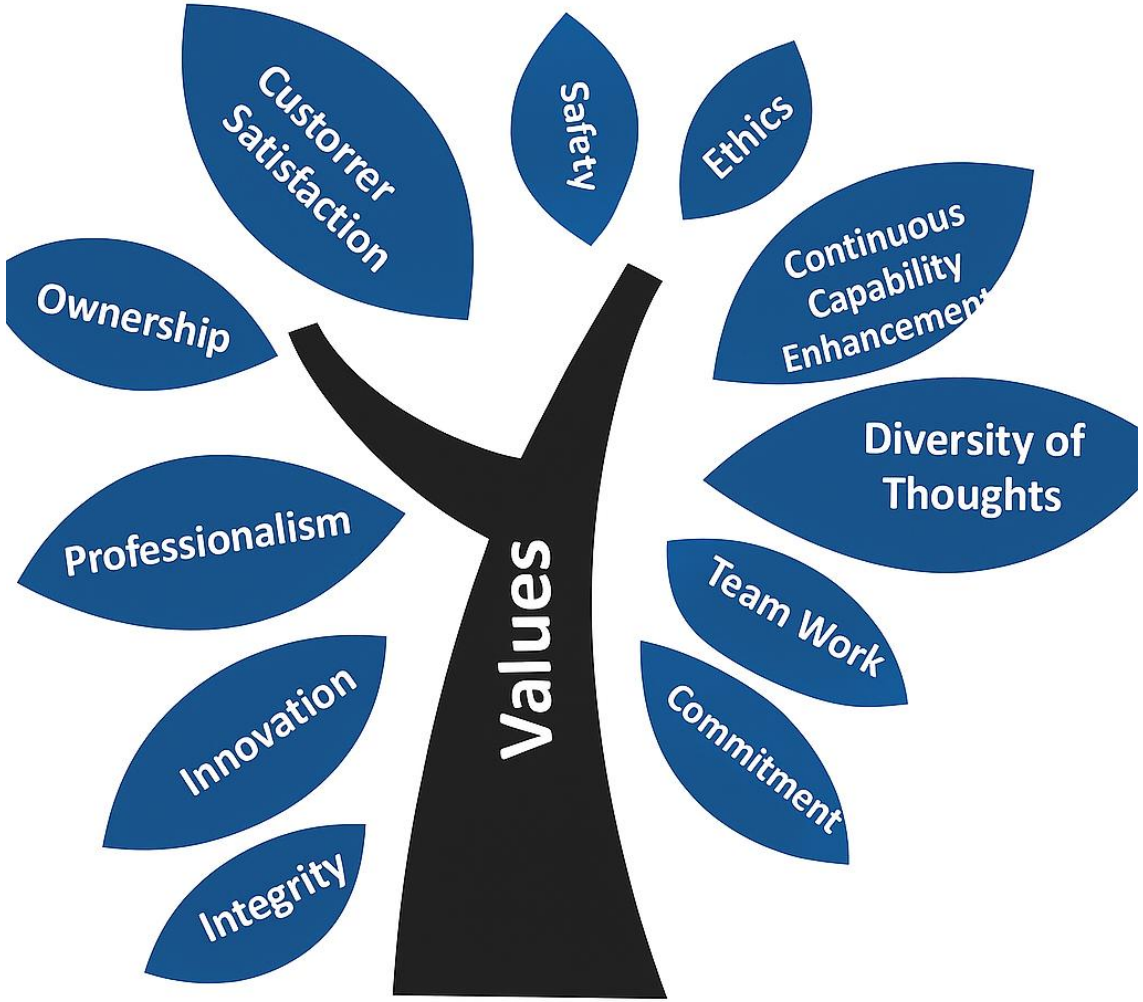
TVS SENSING SOLUTIONS

Automation & IIoT 4.0

www.tvsss.co.in

TVS MOBILITY

Vision, Mission & Values



Vision

"To provide cutting edge automation solutions and resolute focus on customer expectations

Mission

"To help customers with automation through innovation and deploying latest solutions & services"

TVS MOBILITY

TVS SENSING SOLUTIONS

AS TECHNOLOGY RAPIDLY EVOLVES, SO DO WE.



Our Journey

1911
Foundation of TVS Group by TV Sundram Iyengar with focus on transport services

1962
Formation of Sundaram Clayton limited

1994
Formation of EOU to focus on export opportunities

2000
Newer applications with Reed technology

2002
TVS Motor awarded with Deming prize (TVS Group with 5 Deming awards in total)

2005
50:50 JV between TVS & Cherry to manufacture switches

2008
Cherry acquired by ZF; JV renamed ZF Electronics TVS

2012
Identified as engineering support center for ZF global sensor requirements and custom products

2013
Business focus placed on India auto and exports

2018
TVS acquired 100% stake. Renamed as TVS Sensing Solutions Private Limited.

2019
Sales reached all-time high of USD 8.5 bn

2019
FOSS launched

2022
IIoT launched

2023
Electric vehicle products launch

2024
SMT Line installation

TVS

TVS MOBILITY

TVSSS Industry 4.0 Offering

TVS SENSING SOLUTIONS

IIoT

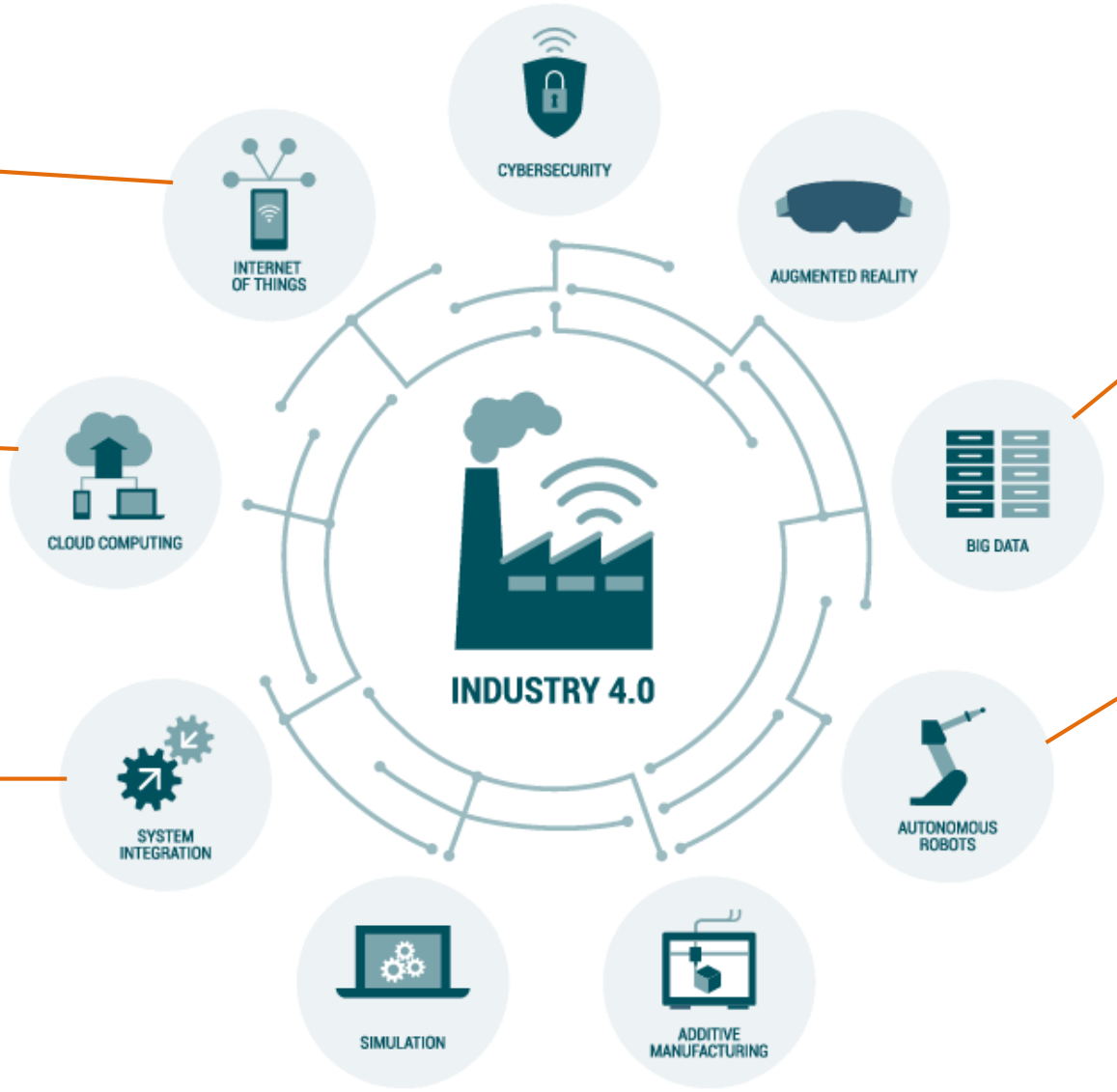
- OEE
- Energy Monitoring
- Real time data capture
- Field data acquisition
- Location Tracking

Cloud Computing

- Any time Anywhere data
- Remote Monitoring and Control
- Notifications / Alerts

System Integration

- Mistake Proofing
- IT / OT Integration
- End-to-end Traceability
- Production Planning & Control
- M2M Communication



Big Data

- Historian
- Genealogy
- *Machine Learning*
- Artificial Intelligence

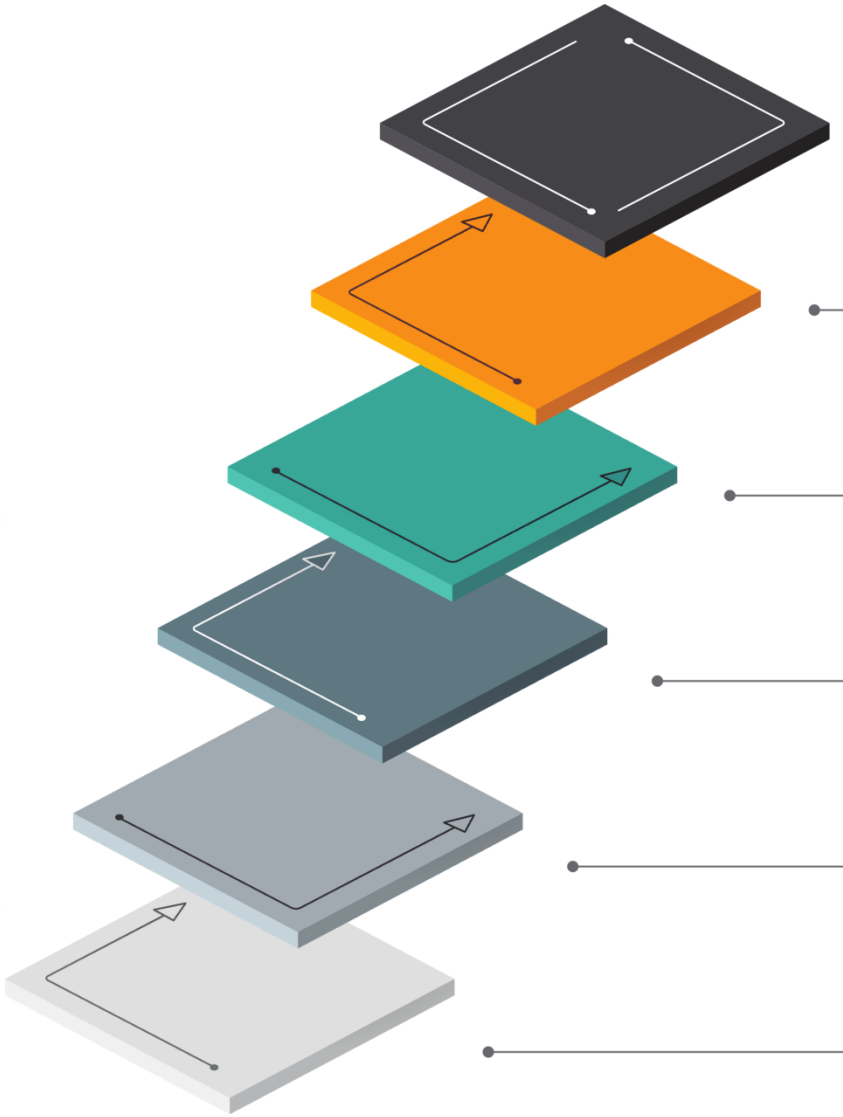
Robotics

- Collaborative Robots
- Articulated/SCARA Robots
- End Effectors
- Turnkey Solutions

TVS MOBILITY

TVS SENSING SOLUTIONS

TVSSS - IIoT Solutions across the Stack



6 Visualization
Web / Mobile, Analytics and Charts, Dashboard, ANDONS

5 Data Warehouse
NOSQL / Timeseries Database, Relational / File Data Store, Data Streams and Warehousing, Machine Learning

4 Cloud Platform
MQTT / REST, Rules Engine, Notifications

3 IoT Gateway
Data Buffering and Aggregation, Cloud / Server Connectivity, Local Decision Making,

2 IoT Nodes
I/O, Data Collection, Signal Control

1 Field Sensors and Instrumentation
Temperature, Pressure, Humidity, Proximity, Switches, Push Buttons, Etc.

TVS MOBILITY

Smart Factory - IIoT Solution Overview

TVS SENSING SOLUTIONS

Cloud Level

Manufacturing Operations Core

Big Data Repository

Cognitive Intelligence

Statistical Process Control

Integrated Maintenance

Collaborative Manufacturing

Plant Level

Sourcing | Supply chain | Manufacturing | Packaging | Logistics | End User Purchase

Industrial Process



Execution Frontend



Plant Monitoring



Notifications and Remote Control

Data Lake

Legacy Equipment

- PLC
- SCADA
- OPC Server

Greenfield Equipment

- M2M Communication
- Machinery Sensor data
- Machine Process data

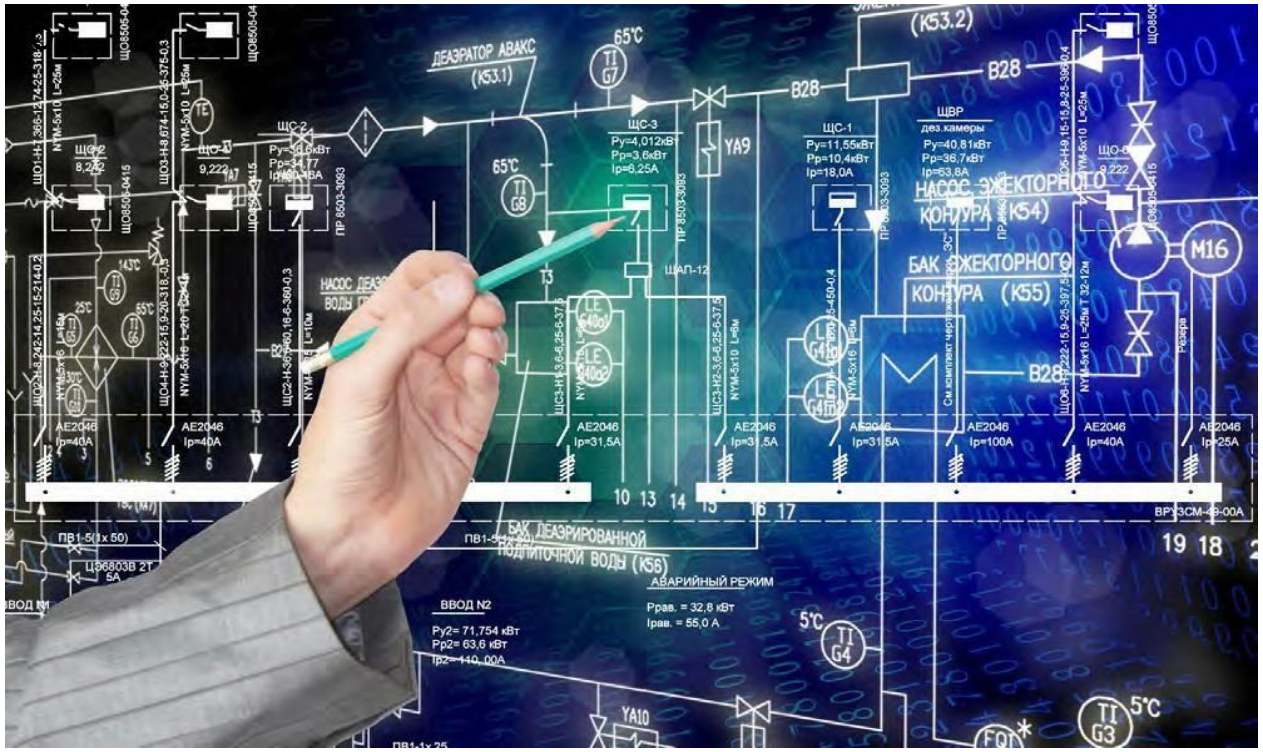
Environmental Sensors

- IoT Sensor Network (Temperature, Pressure, Gas, Noise, Vibrations etc.)

TVS MOBILITY

IIoT Solutions Core

TVS SENSING SOLUTIONS



- Electrical Design & Drafting
- PLC/SCADA Program Development
- Robot Programming & Simulations
- Production Monitoring & Supervision
- Instrument Configuration & Installation
- Field Engineering
- Mechanical Engineering
- Analysis & Simulation

Specialization

E-Plan	NI Lab View Automation	Robot Studio
Auto Cad	Special Purpose Machine(SPM)	EOL Automated Test Equipment’s
Solid Works	Special Test Rigs / Test Benches	Traceability & Realtime Statistical Analysis Industrial Software

TVS MOBILITY

TVS SENSING SOLUTIONS

IIoT- Serviceable Business Verticals



Automotive



Tyre Industry



FMCG



Energy & Utilities



Food Industries



Textile Industry



Electronics Industry



Supply Chain Industry



Pharmaceutical



Other manufacturing

TVS MOBILITY

IIoT – Process driven solutions

TVS SENSING SOLUTIONS

Line Automation (Assembly Line)



Robotic Solutions



Special Test Rigs / Test Bench (SPM / EOL)





Conveyor Solutions



Optical Vision Solutions



TVS MOBILITY

IIoT – Process driven solutions

TVS SENSING SOLUTIONS

Traceability Solutions



Smart Bin Solutions



Energy Management



Independent Stations



Portable Test Kit



Digital Solutions



TVS MOBILITY

Line Automations (Assembly Line)

TVS SENSING SOLUTIONS









Concept Design
- ✓ 3D Designing
- ✓ 2D Detailing

Validation
- ✓ Feasibility study
- ✓ Verification

Manufacturing
- ✓ Fabrication
- ✓ Assembly

Testing
- ✓ Integration
- ✓ Trial Running

Commissioning
- ✓ Installation of system
- ✓ Prove out at site

TVS MOBILITY

Robotic Solutions

TVS SENSING SOLUTIONS



- Machine Tending
- Material Handling
- Dispensing & Deburring
- Foundry & Forging
- Palletizing & Part Transfer
- Inspection – CMM/Vision

Concept Design

- ✓ 3D Designing
- ✓ 2D Detailing

Simulation

- ✓ Feasibility study
- ✓ Cycle-time study

Manufacturing

- ✓ End of Arm Tooling
- ✓ Mechanical System

Interfacing

- ✓ Control System
- ✓ Programming

Commissioning

- ✓ Installation of system
- ✓ Site prove out

TVS MOBILITY

Special Test Rigs / Test Bench (SPM / EOL)

TVS SENSING SOLUTIONS



- NI LabVIEW Automation
- Special Purpose Machine (SPM) - Customized
- Special Test Rigs / Test Benches - Customized
- End Of Line (EOLT) Automated Test Equipment's
- PLC/SCADA/LabVIEW Program Development
- Traceability & Realtime Statistical Analysis

Industrial Software

Design
- ✓ 3D Designing
- ✓ 2D Detailing

Validation
- ✓ Feasibility study
- ✓ Verification

Manufacturing
- ✓ Fabrication
- ✓ Assembly

Testing
- ✓ Integration
- ✓ Trial Running

Commissioning
- ✓ Installation of system
- ✓ Prove out at site

TVS MOBILITY

Conveyor Solutions

TVS SENSING SOLUTIONS



- Roller Conveyors
- Chain Conveyors
- Pallet Conveyors
- Belt Conveyors
- Elevating Solutions
- Balancing Solutions
- Structural Systems
- Customized Conveyor Solutions

Design
- ✓ 3D Designing
- ✓ 2D Detailing

Validation
- ✓ Feasibility study
- ✓ Verification

Manufacturing
- ✓ Fabrication
- ✓ Assembly

Testing
- ✓ Integration
- ✓ Trail Running

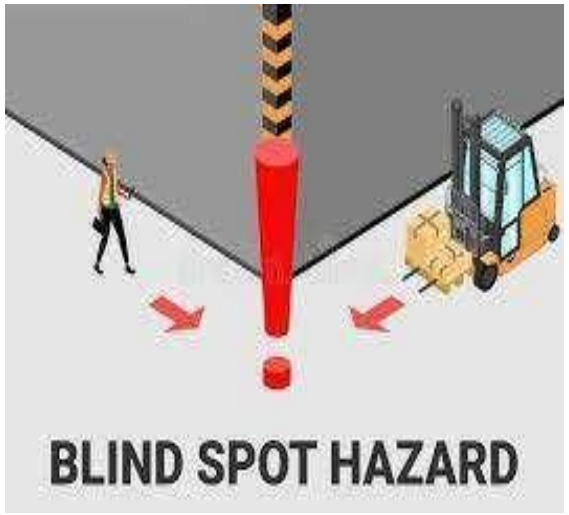
Commissioning
- ✓ Installation of system
- ✓ Prove out at site

TVS MOBILITY

Optical Solution (Safety Solutions with Optical Vision)

TVS SENSING SOLUTIONS

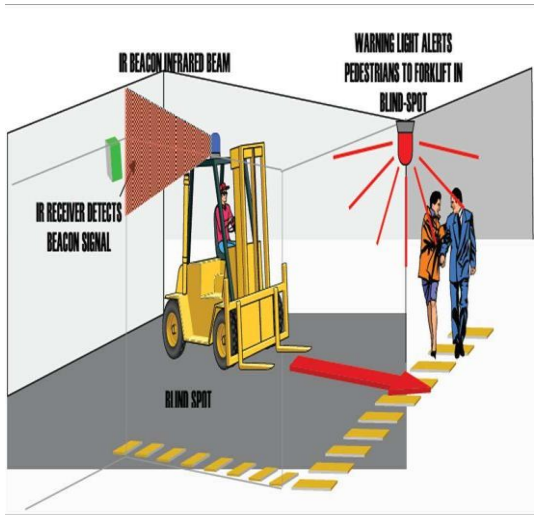
Man – Vehicle Collision :



Man – Vehicle Collision Points:



Sign Projectors – STOP



Alert System – Indication with Buzzer

Man – Vehicle Collision :
Safety system detects the Man and vehicle, “STOP” sign signal projector for floor display. This System detect possibility of collision gives alert Light with buzzer.

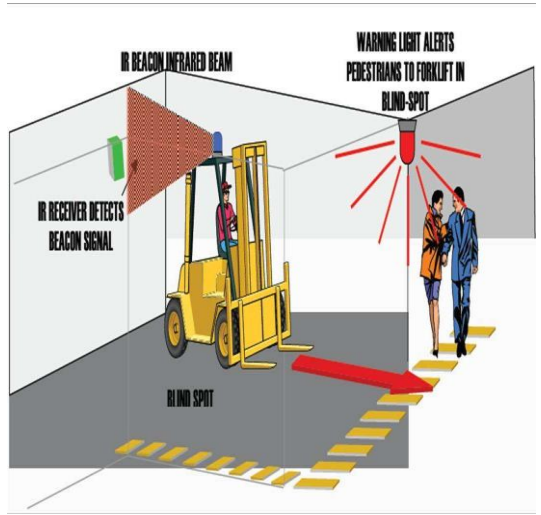
Vehicle – Vehicle Collision :



Vehicle – Vehicle Collision Points:



Sign Projectors – STOP



Alert System – Indication with Buzzer

Vehicle – Vehicle Collision :
Safety system detects the Vechicle and vehicle, “STOP” sign signal projector for floor display. This System detect possibility of collision gives alert Light with buzzer.

TVS MOBILITY

Traceability solutions:



- ✓ Track and Trace Solutions (Barcode & RFID)
- ✓ Mistake Proofing / Poka Yoke / Digital Walls and Dashboards
- ✓ Real Time Manufacturing information (Machine Monitoring /OEE /Energy Monitoring)
- ✓ Handheld and Mobility Solutions
- ✓ ERP and Third Party Integration / Warehouse and Stores Management

TVS SENSING SOLUTIONS

Smart Bin solutions:

- ✓ Smart bins can significantly lower operational costs.
- ✓ Real-time data / Timely alerts and optimized collection prevent overflowing bins.
- ✓ Smart bins provide valuable data on waste generation patterns, enabling better planning and resource allocation.
- ✓ Smart bins have safety features like sensors that prevent compaction if a hand is detected or alert authorities in case of fire.



Energy Management:



- ✓ Cost Savings
- ✓ Improved Efficiency
- ✓ Sustainability and Environmental Impact
- ✓ Increased Return on Investment (ROI)
- ✓ Improved Compliance
- ✓ Data-Driven Decision Making

TVS MOBILITY

Independent stations

TVS SENSING SOLUTIONS



AI Vision System - Hydraulic Cylinder Pressing Machine



Performance Test Machine



ABS Wheel Speed Sensor



Roller Jamming Machine – Tyre Industry:



Welding Machine



IC Forming Machine



Servo Based Pressing Machine



4 Axis Robot

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Portable Test Kit

TVS SENSING SOLUTIONS

Gear Position Sensor (GPS) – Test Kit



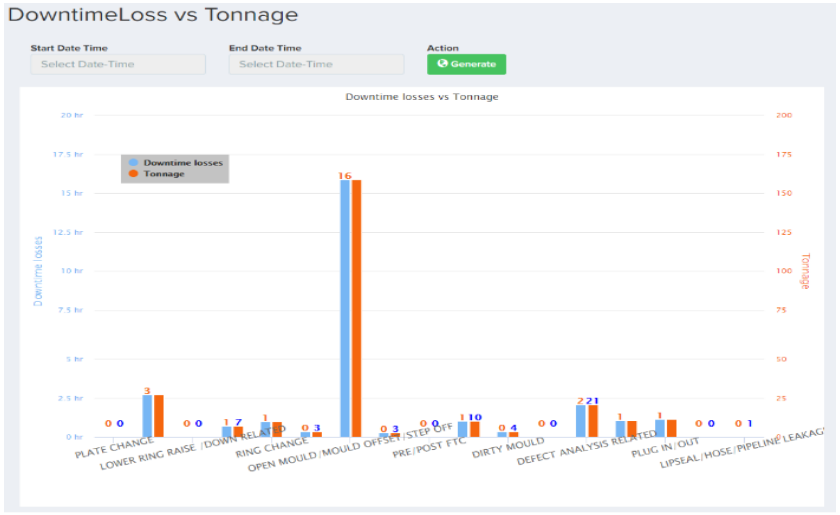


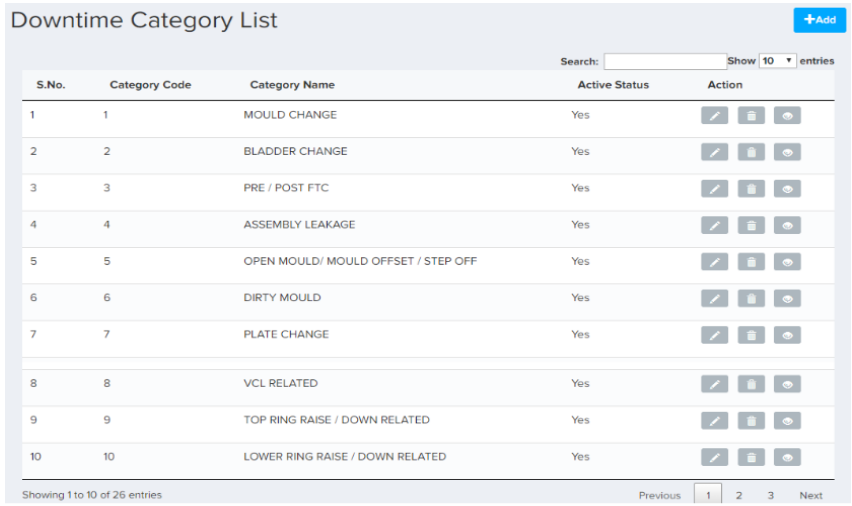
TVS MOBILITY

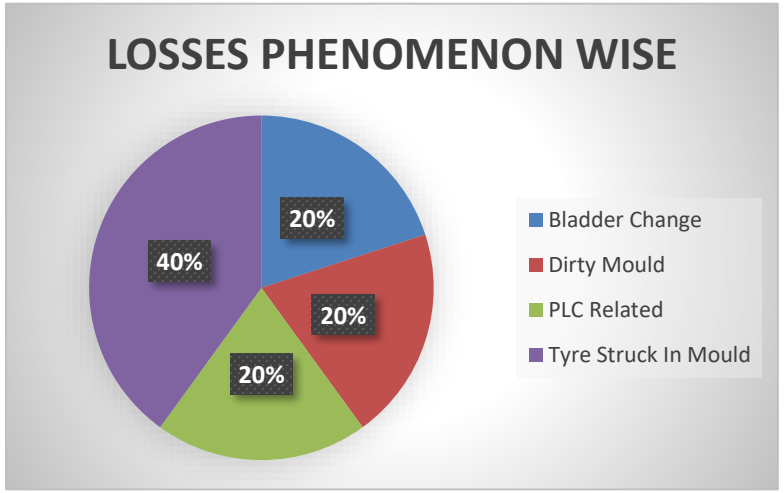
Dashboards (Digital Solutions)

TVS SENSING SOLUTIONS

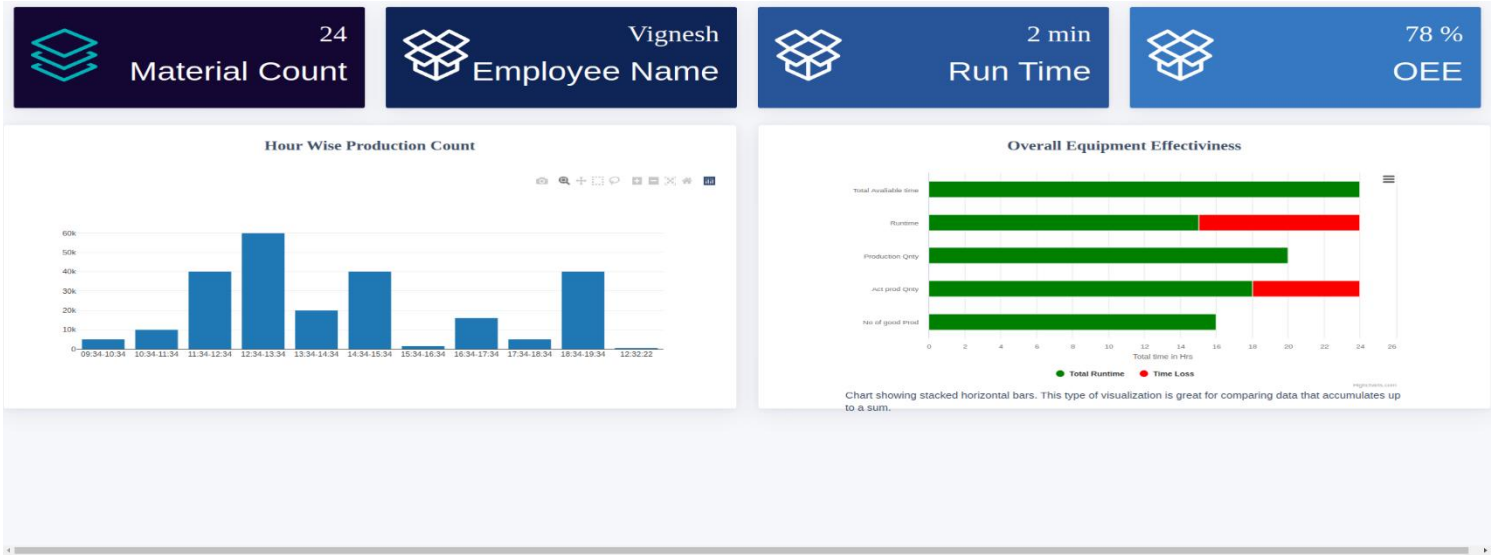
Loss Categorization





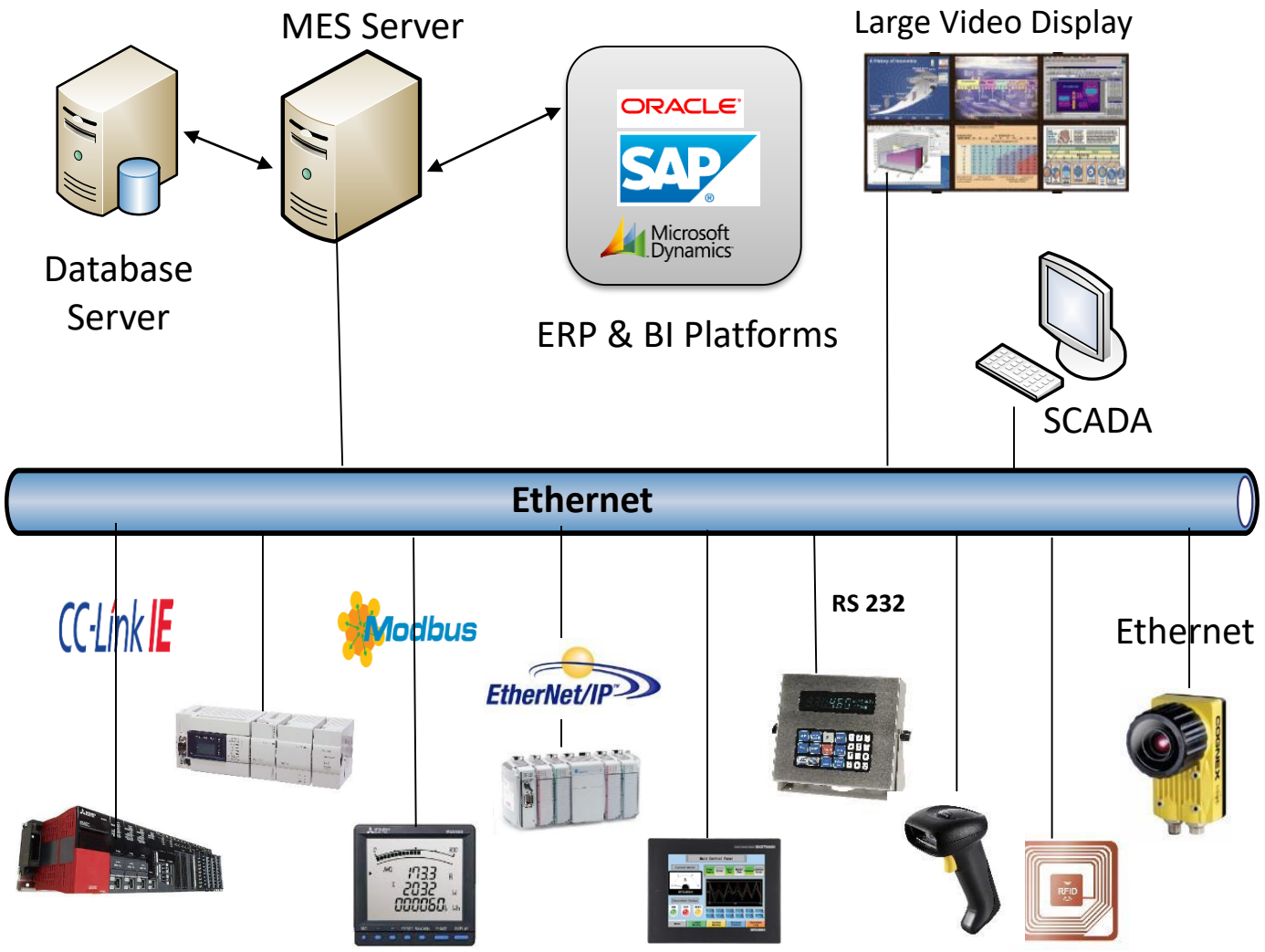


OEE Monitoring



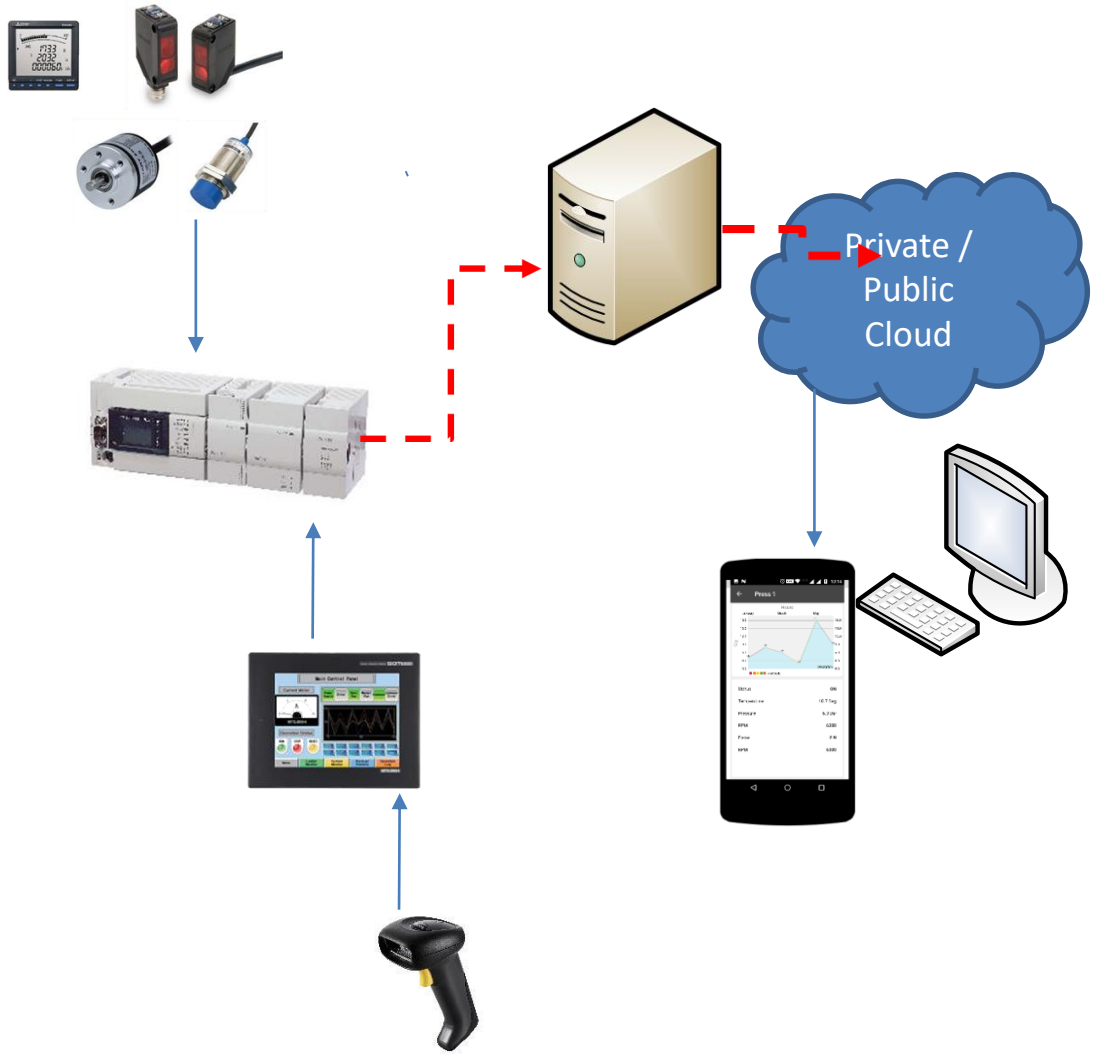
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Typical OT Network



Typical Architecture

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Partners

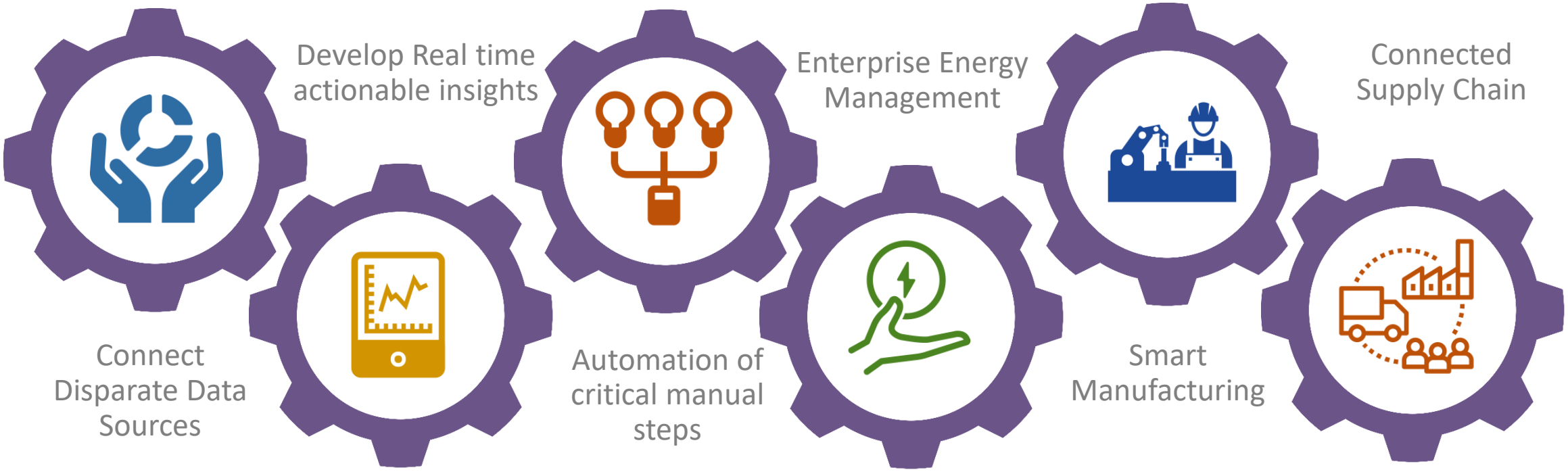
TVS SENSING SOLUTIONS

Robot	PLC / SCADA	Mechanical /Electrical
ABB	Siemens , ABB	Solid works
Fanuc	Mitsubishi, Omron	AutoCAD
Motoman	Allen Bradly (Rockwell)	Draft Sight
Epson	Movicon, Schneider	CATIA,AUTODESK INVENTEOR
Mitsubishi	National Instruments – Lab View	e PLAN

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From Efficiency to Transformation

TVS SENSING SOLUTIONS



Productivity & Efficiency Potential

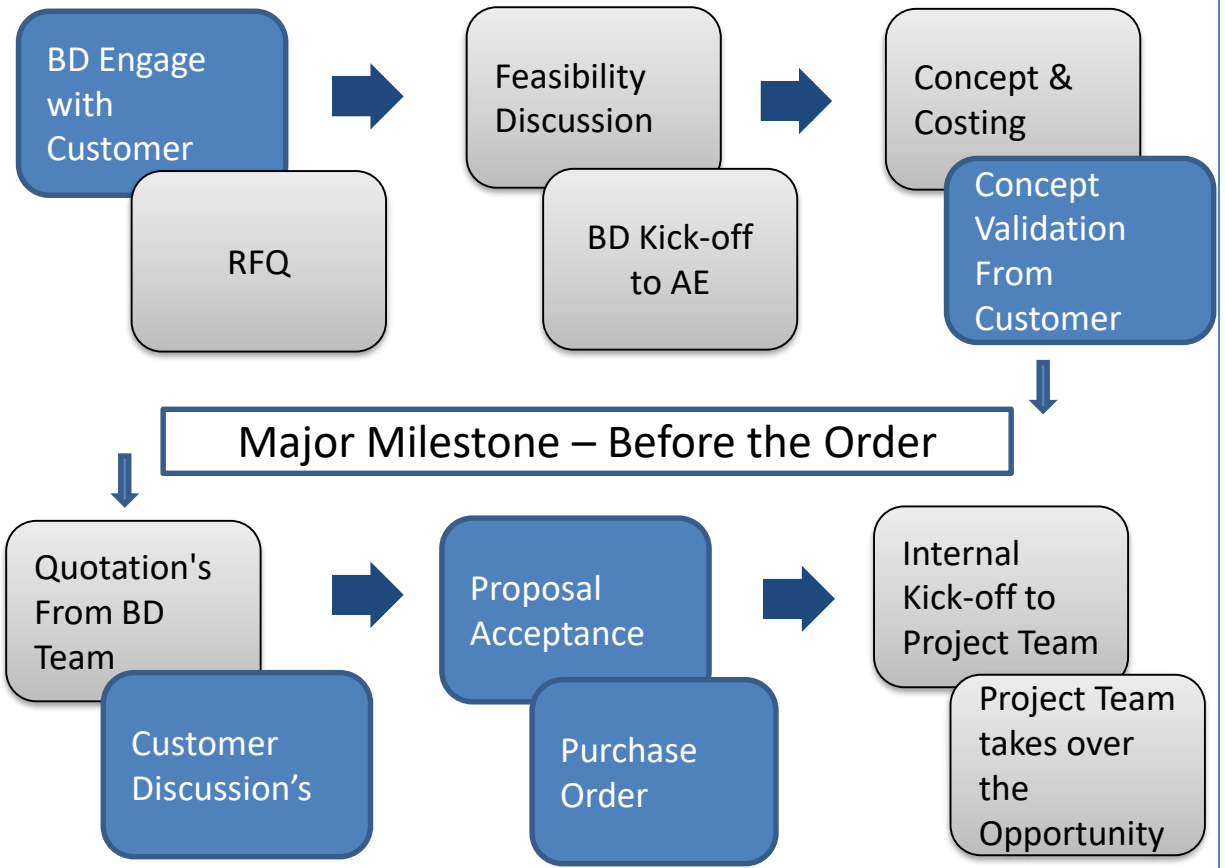
Transformative Potential

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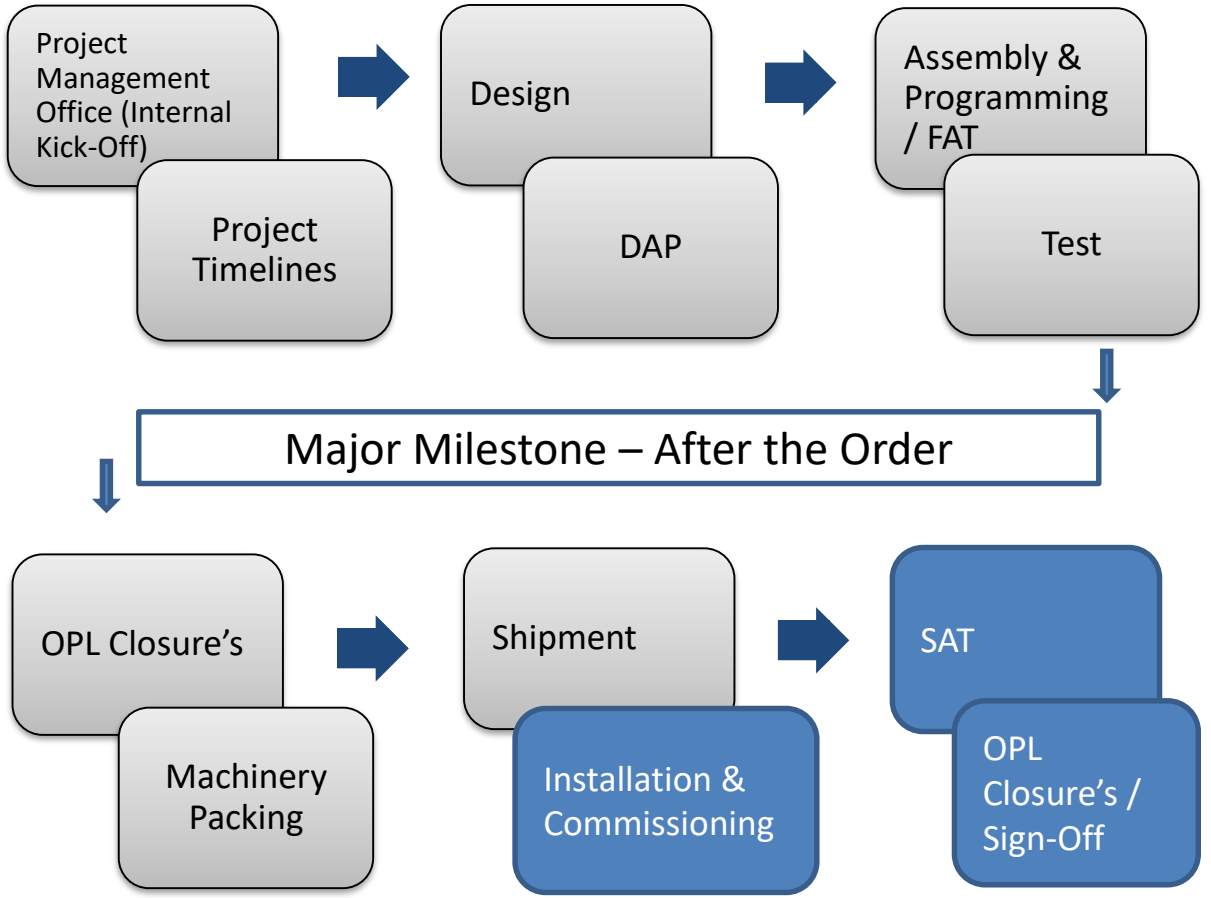
Project Flow

TVS SENSING SOLUTIONS

RFQ Stage



Implementation Stage



External Process | Internal Process

External Process | Internal Process

TVS MOBILITY

Few of our major customers

TVS SENSING SOLUTIONS

wipro

JK Fenner

DELPHI-TVS Diesel Systems Limited

ROYAL ENFIELD

TVS RUBBER

TVS AIR SPRINGS

TVS EUROGRIP

Brakes India Private Limited
Brakes

TVS ARGOMM

CapGrid
reimagining strategic sourcing

DAIMLER

UCAL FUEL SYSTEMS

TEL
TURBO ENERGY

TVS MOBILITY

Contact us:

THANK YOU!

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